# Rahul Nair

Thiruvananthapuram, Kerala | +91 90000 22222 | rahul.nair.robotics@example.com |
linkedin.com/in/rahul-nair-demo | github.com/rahul-ros-demo

## PROFESSIONAL SUMMARY

Robotics Software Engineer with 3 years of experience building autonomous mobile robots using ROS 2. Experienced in navigation, SLAM, sensor fusion and simulation in Gazebo. Strong C++ and Python developer.

## TECHNICAL SKILLS

**Robotics:** ROS, ROS 2 (Humble), Nav2, MoveIt, Gazebo, RViz, SLAM (Cartographer, SLAM Toolbox), URDF
**Programming:** C++, Python, Bash
**Perception:** OpenCV, Computer Vision, Point Cloud Library (PCL), LiDAR, Depth Cameras
**Tools:** Linux (Ubuntu), Git, Docker, CI/CD with GitHub Actions
**Hardware:** Raspberry Pi, NVIDIA Jetson Nano, Arduino, IMU, Motor Controllers
**Concepts:** Kalman Filters, PID Control, Path Planning, Kinematics

## EXPERIENCE

**Robotics Software Engineer, AutoBots Technologies, Technopark Trivandrum | Aug 2023 - Present**

- Developed ROS 2 navigation stack for a warehouse robot fleet of 15 robots using Nav2 and SLAM Toolbox.
- Implemented sensor fusion of LiDAR, IMU and wheel odometry with an Extended Kalman Filter, improving localization accuracy by 30%.
- Built Gazebo simulation worlds and automated tests in Docker, cutting field-testing time by 40%.
- Mentored 2 interns on ROS 2 and C++ best practices.

**Junior Robotics Engineer, Kerala Robotics Lab, Kochi | Jun 2022 - Jul 2023**

- Programmed a 6-DOF robotic arm with MoveIt for pick-and-place of packages.
- Wrote OpenCV-based object detection to locate items on a conveyor belt at 25 FPS on Jetson Nano.

## PROJECTS

**Autonomous Campus Delivery Robot | ROS 2, Nav2, Python**

- Built a delivery robot that mapped a 2 km campus and delivered parcels with 95% success across 200 test runs.

**Line-Following Drone Simulation | Gazebo, PX4, C++**

- Simulated a drone following power lines using camera input and a PID controller.

## EDUCATION

**B.Tech in Electronics and Communication Engineering, College of Engineering Trivandrum | 2018 - 2022**

- CGPA: 8.1 / 10

## ACHIEVEMENTS

Winner, Kerala Startup Mission Robotics Challenge 2021
Contributor to open-source ROS 2 packages (navigation plugins)